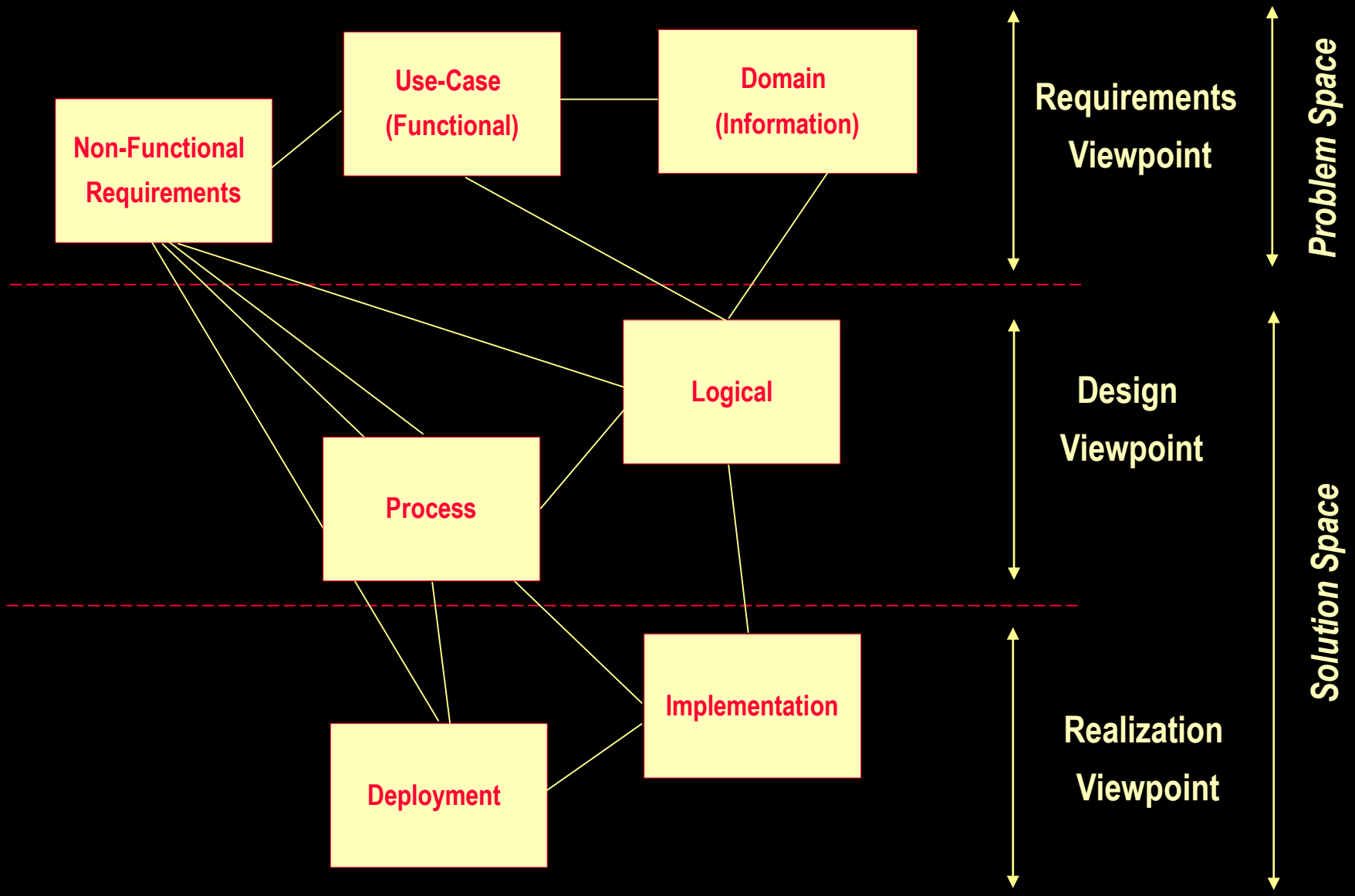
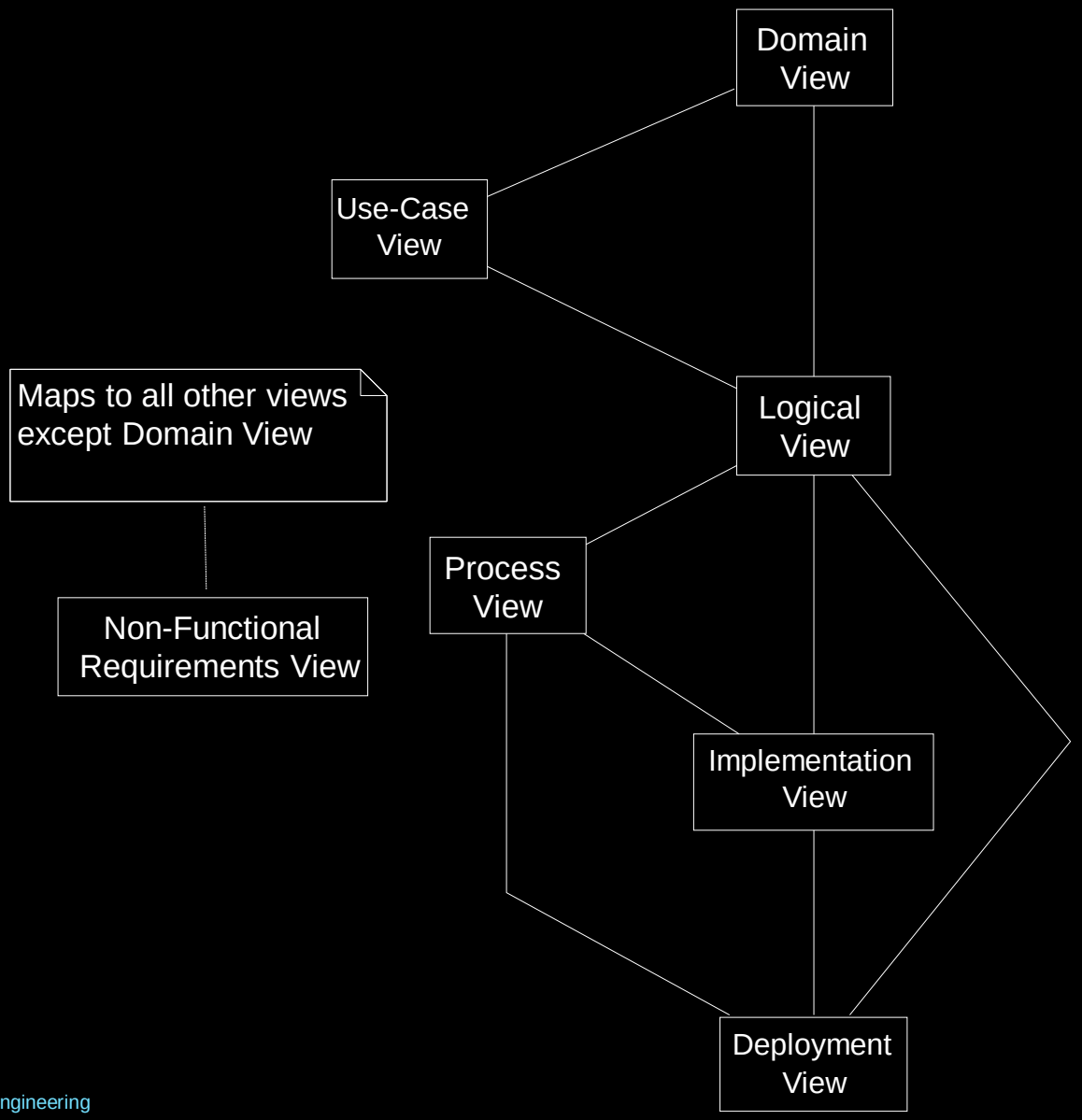
Review: Proposed Architectural Viewpoints and Views



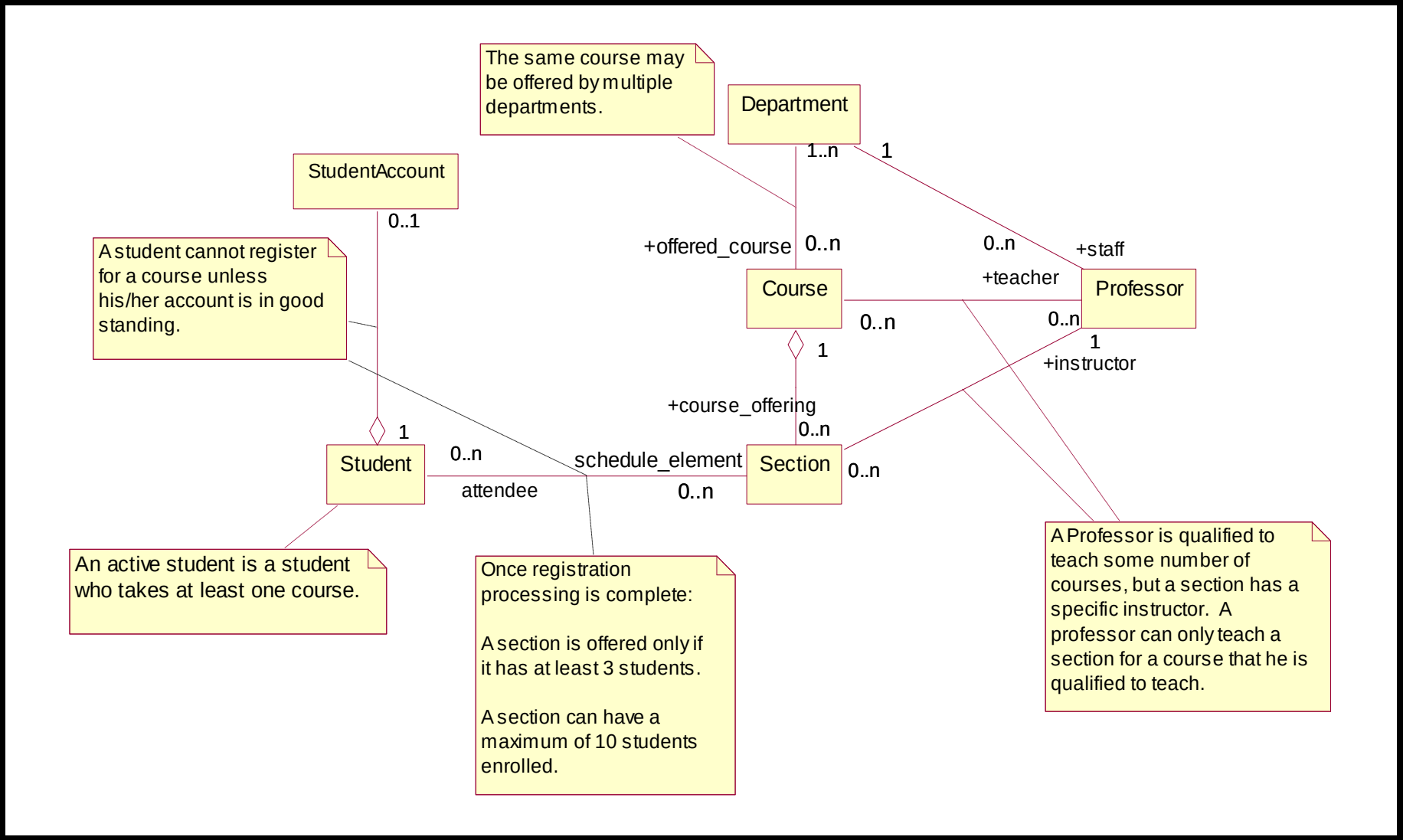
Mappings Between the Views



Domain View Representation

- Structure
  - Class diagrams
- Behavior
  - Interaction diagrams
  - Statechart diagrams
  - Activity diagrams
  - Object Constraint Language (OCL)

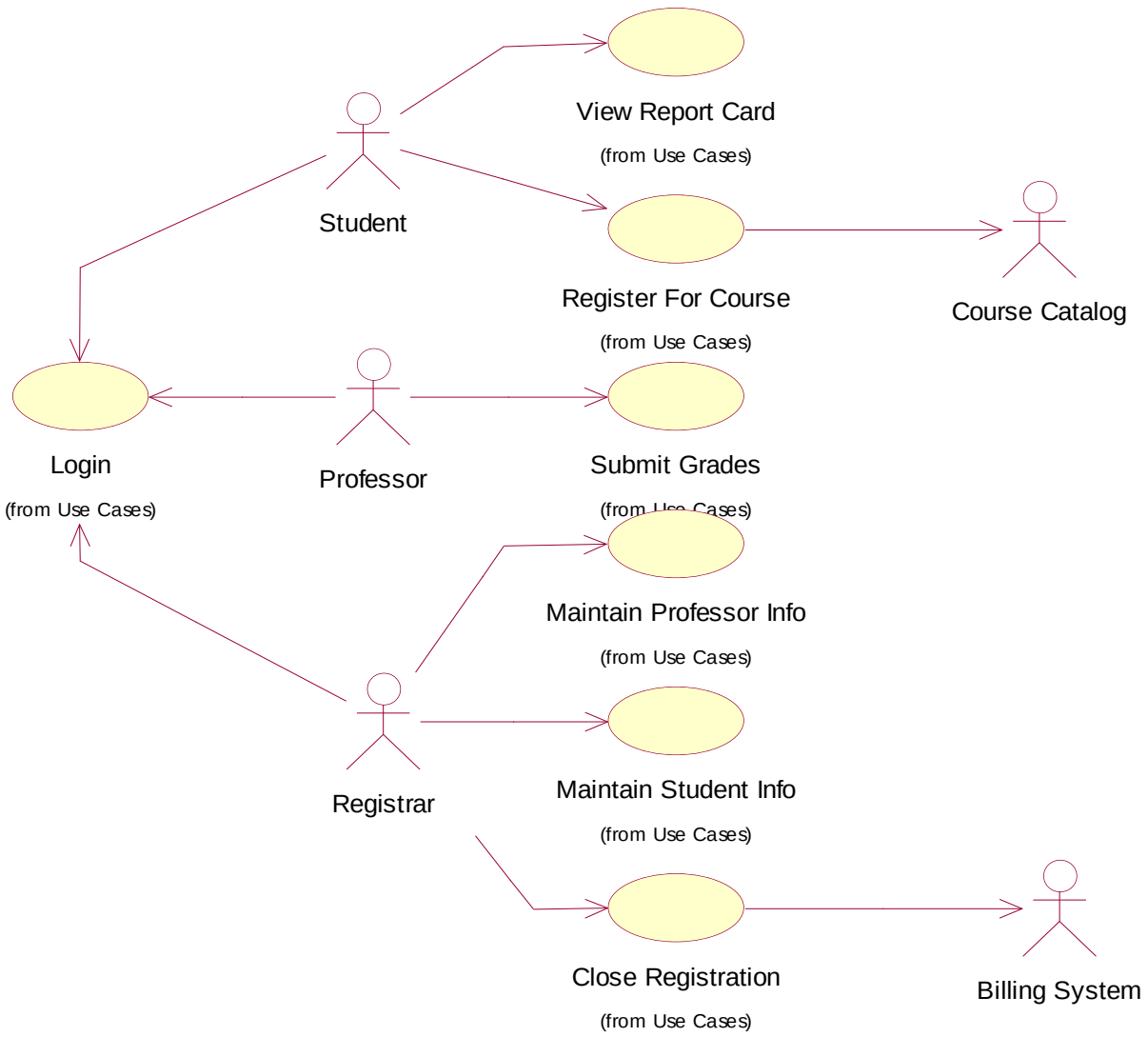
Example: Domain View: Class Diagram



Use-Case View Representation

- Structure
  - Use-case diagrams
- Behavior
  - Use-case textual descriptions
  - Use-case activity diagrams
  - Use-case interaction diagrams (sequence or collaboration)
  - Textual description of how the requirements can change over time and how the architecture has been designed to cope with this

Example: Use-Case View: Use-Case Diagram



Example: Use-Case View: Textual Description

Register For Course.doc - Microsoft Word

File Edit View Insert Format Tools Table Window Help

1. Register for course over the Internet

The use case describes how a student can register for an offered course using the university Web site.

2. Pre and Post Conditions

Pre-conditions: - the student has logged into the system and his/her identity has been verified

Post-conditions: - student has been signed for a section of the selected course and the Billing System has been informed to record tuition charge

3. Flow of events

3.1 Basic flow

#	Event	Comments
1	Student selects course registration page on the university Web site	
2	Student is show a course search page to describe course he/she is interested in	Courses are organized by departments, then by major and then by level (100, 200, etc.)
3	Student is asked to select a course for the presented list	
4	The systems shows open section of the selected course	If there are no open sections, the student is informed about that and shown the course search page again
5	Student selects a section	
6	The system shows student date and asks for confirmation	The system stores student data during the login and can display them at any time during a session
7	The system displays a conformation	
8	A payment record is sent to the Billing System	Payment records are accumulated until the end of the sing-up session
9	End use-case	

3.2 Alternative flow(s)

1. The is a similar use case for registering over the local area network

4. Special Requirements

Student can stop the registration process at any time.

Registration is not committed until step 7 in the use case (the confirmation)

5. Extension Points

None

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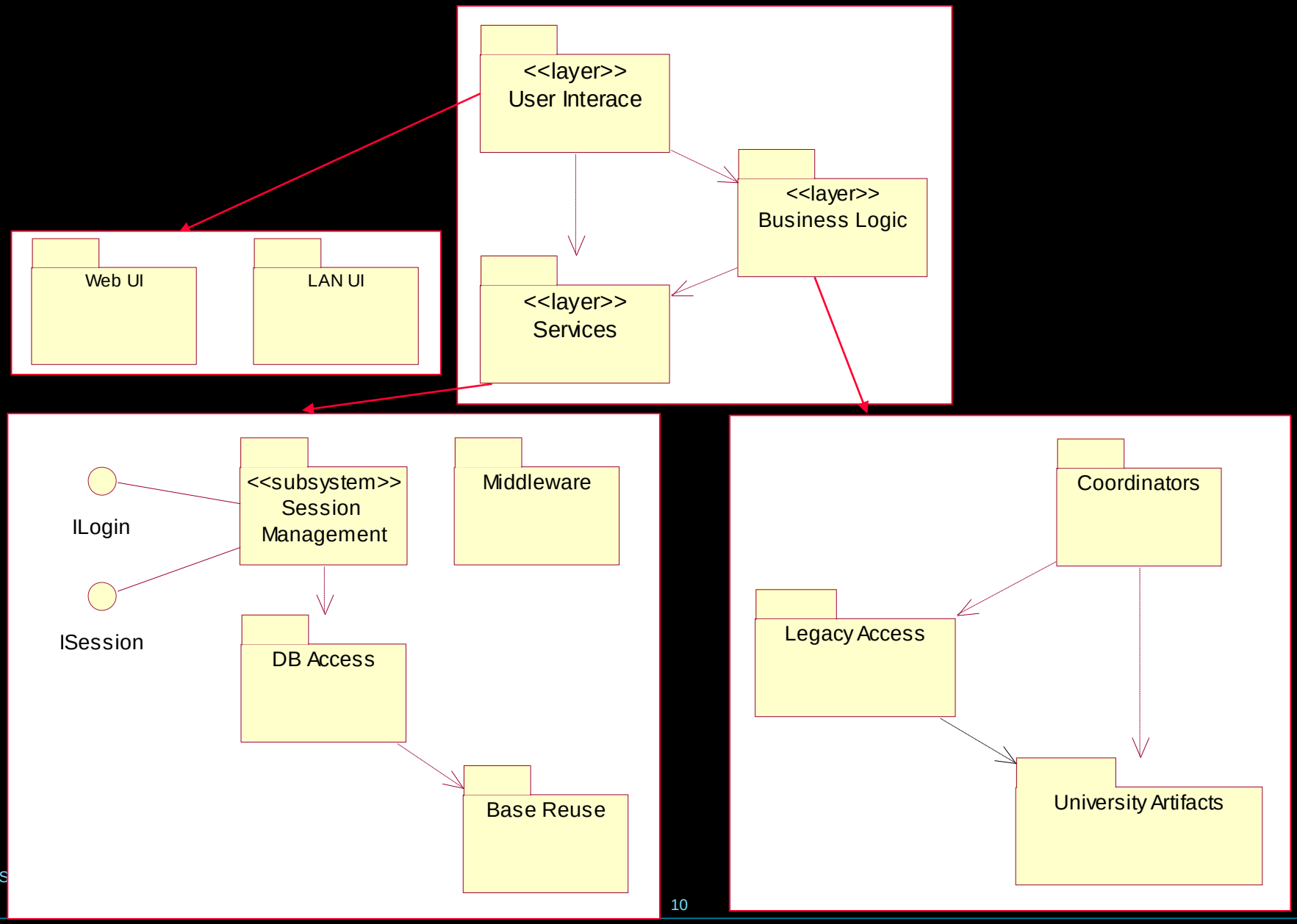
Example: Non-Functional Requirements View

Requirement	Question	Impact	Answer	Priority
The system should provide both the Internet and LAN access	Why not use only Web-based access?	Will require developing two alternative interface paths to the system. Likely to increase the system cost by 25%	We must have dedicated, high-speed access to the system.	High
The system should use Oracle 8 running on Sun/Solaris	Why not allow the data to be stored on SQL Server running under Win NT 4.0	Must have people with both NT and Unix/Solaris skill on the team	The Catalog system uses Oracle and we want to support only one DB server and one DBMS	High
The middle tier of the C/S architecture must run on Windows NT 4.0 servers	If we are using Oracle on Sun, why not use Sun servers to develop the middle tier?	By using one technology family we could save 15% on development costs	Our long-tem plan is to migrate our system to MS platform.	High
The system should process at least 7 course registrations per second			= 100/15 → 100 users and 15 sec. maximum response time)	Medium
….				

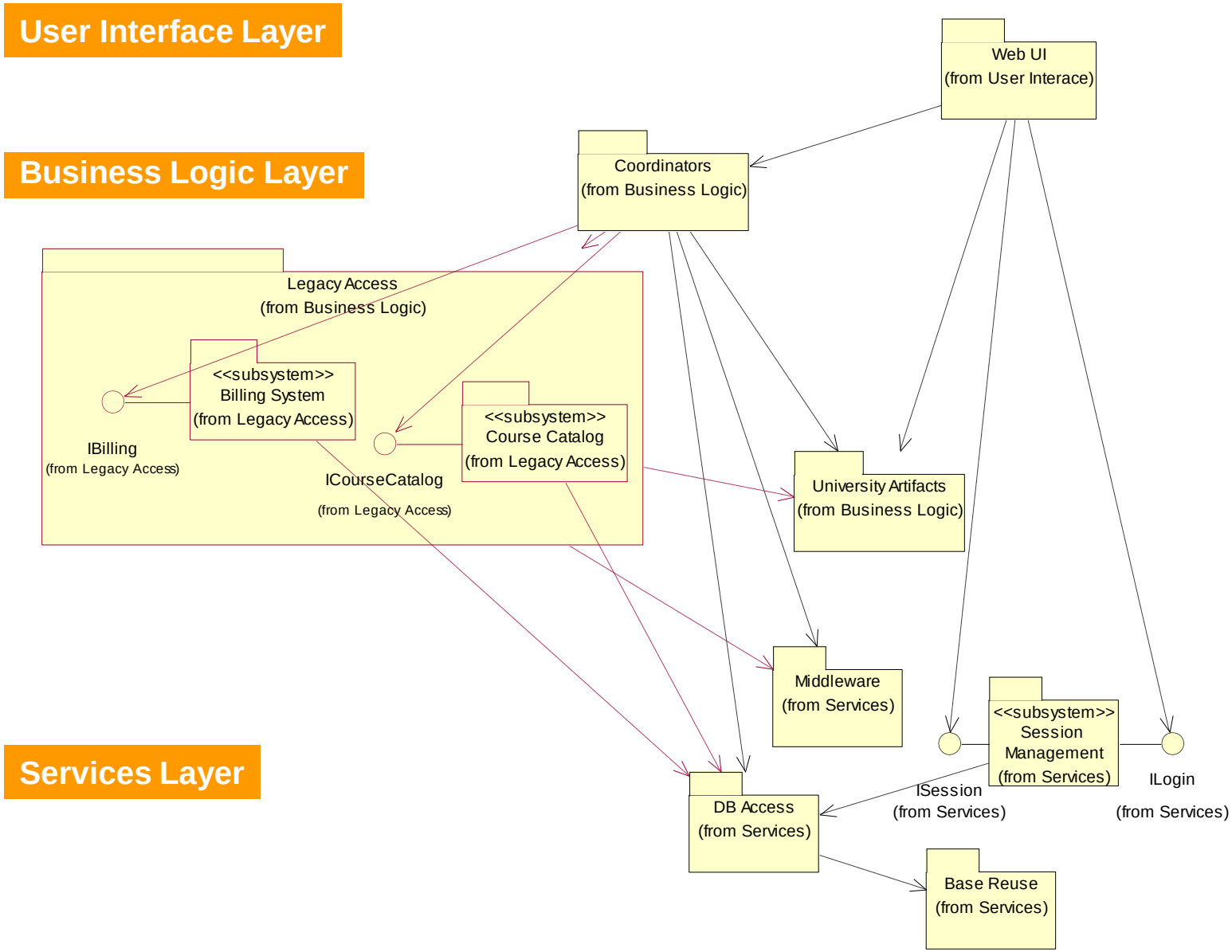
Logical View Representation

- Structure
  - Package Diagrams
    - Major partitions (packages)
  - Class Diagrams
    - Architecturally-significant design elements (classes, subsystems, interfaces) and their relationships
- Behavior
  - Interaction diagrams (sequence and collaboration)
  - Statechart diagrams
- Collaborations
  - Contain both a structural and a behavioral aspect

Example: Logical View: High-Level Structure

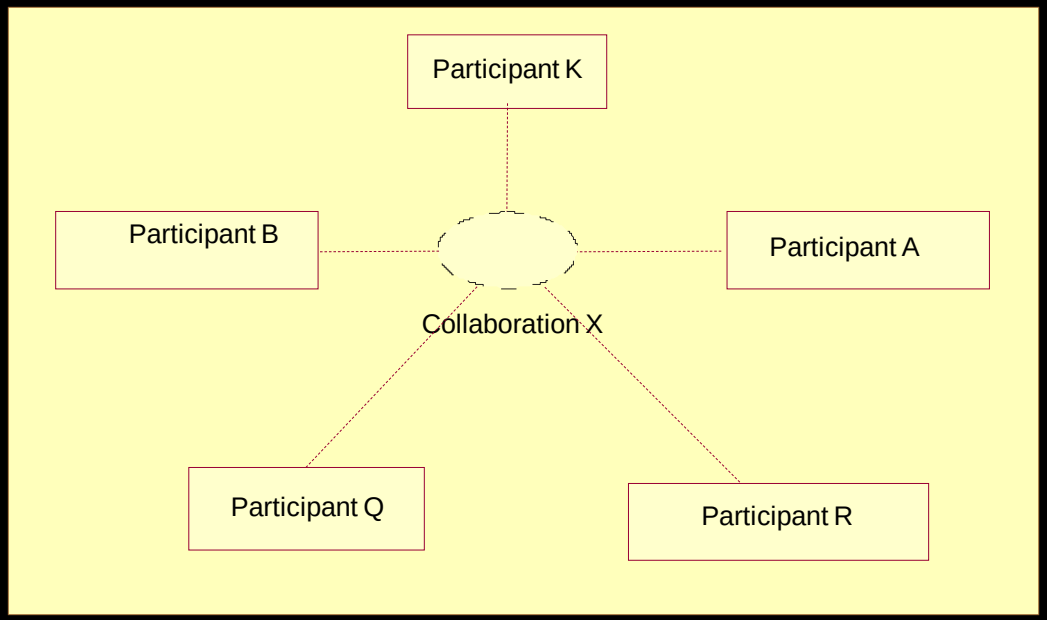


Example: Logical View: Package and Subsystem Layering



Logical View: Collaboration

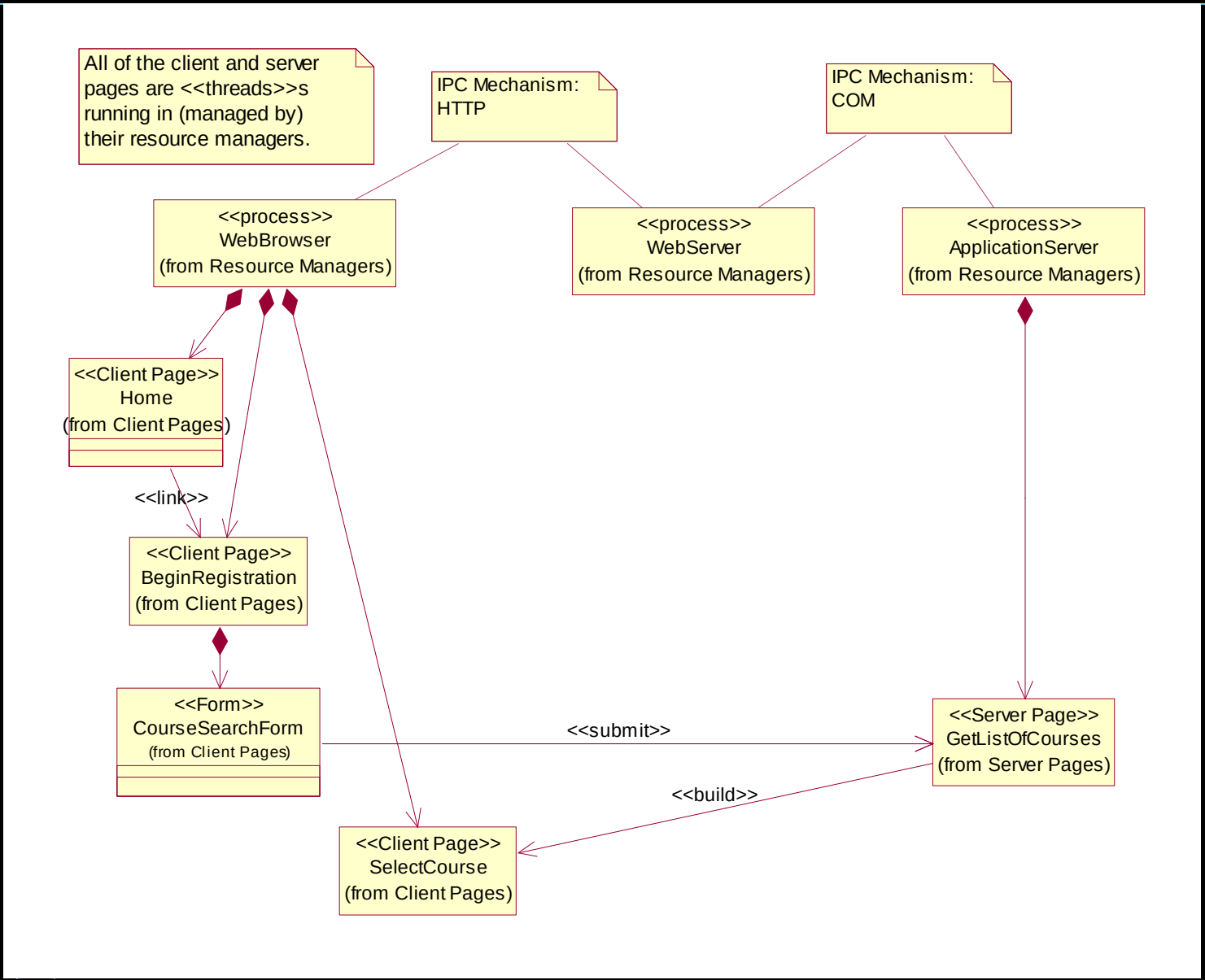
- Groups elements working together to realize a specific system behavior
  - Mechanism
  - Use-case (use-case realization)
- Has static and dynamic parts/facets



Process View Representation

- Structure (cont)
  - Object Diagrams (run-time image)
    - Active class instances
    - Communication links
  - (at least) One object diagram per process
- Behavior
  - Interaction diagrams (sequence or collaboration) showing IPC
    - Active instances
    - Communication
  - Activity diagrams
  - At least one diagram per process

Example: Process View: Class Diagram



Component Categories

- Work product components
  - The source code files, data files, etc. used to create the deployment components
  - The physical packaging of logical/conceptual things (e.g., classes, processes, etc.)
  - Examples: .h, .cpp and .hpp files for C++, CORBA IDL, or.java files for Java, build files (makefiles)
  - Supports visualization of the compilation dependencies among the source code elements (important when managing the configuration management and version control)

Component Categories (2)

- Deployment components
  - The parts needed to deliver a running system
  - Include executables and libraries (things that execute), as well as ancillary components (that don't execute)
  - Examples: EXE files, DLLs, COM stuff, JARs, database tables, data files, help files, scripts, initialization files, log files, applets (.class files), web pages (.htm and .html files)
  - Supports visualization of the (physical) configuration of the deployment components (important for deployment and configuration management)

Component Categories (3)

- Execution components
  - Physical things created and destroyed dynamically as the application runs
  - Examples: COM+ objects, temp files, server/servlet generated web pages, DHTML, XML messages sent between processes

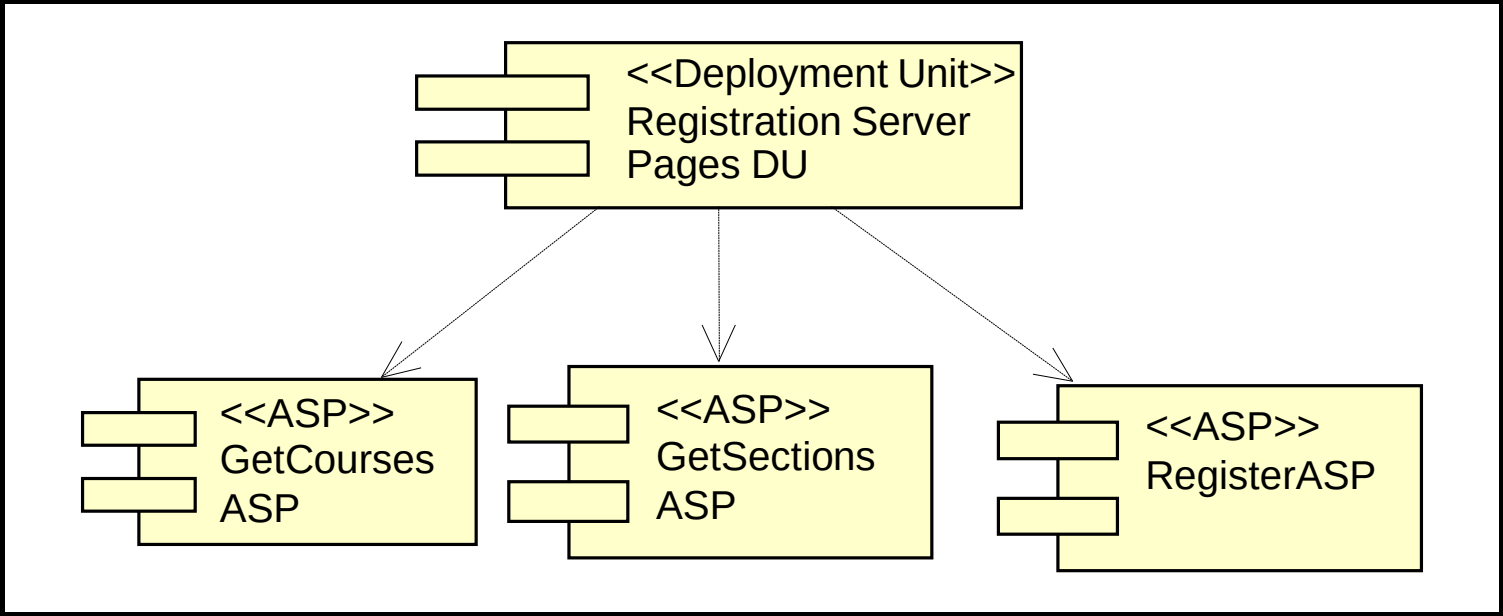
Implementation View Representation

- Component Diagrams
  - Work product, deployment, and execution components
  - Deployment units
  - Stereotyped components (e.g., executable, library, table, file, document, deployment unit)
  - Dependencies between components
    - From deployment components to the work product components they are built from
    - From deployment components to the execution components created when the deployment components run
    - From deployment units to the deployment components that will be deployed in them

Implementation View Representation (2)

- Component Diagrams (cont.)
  - Interfaces and their realization relationships with components
  - Packages
    - Logical groupings/partitions
  - Component diagram(s) for each deployment component
    - Showing associated work product components
    - Showing associated execution components
  - One component diagram for each deployment unit

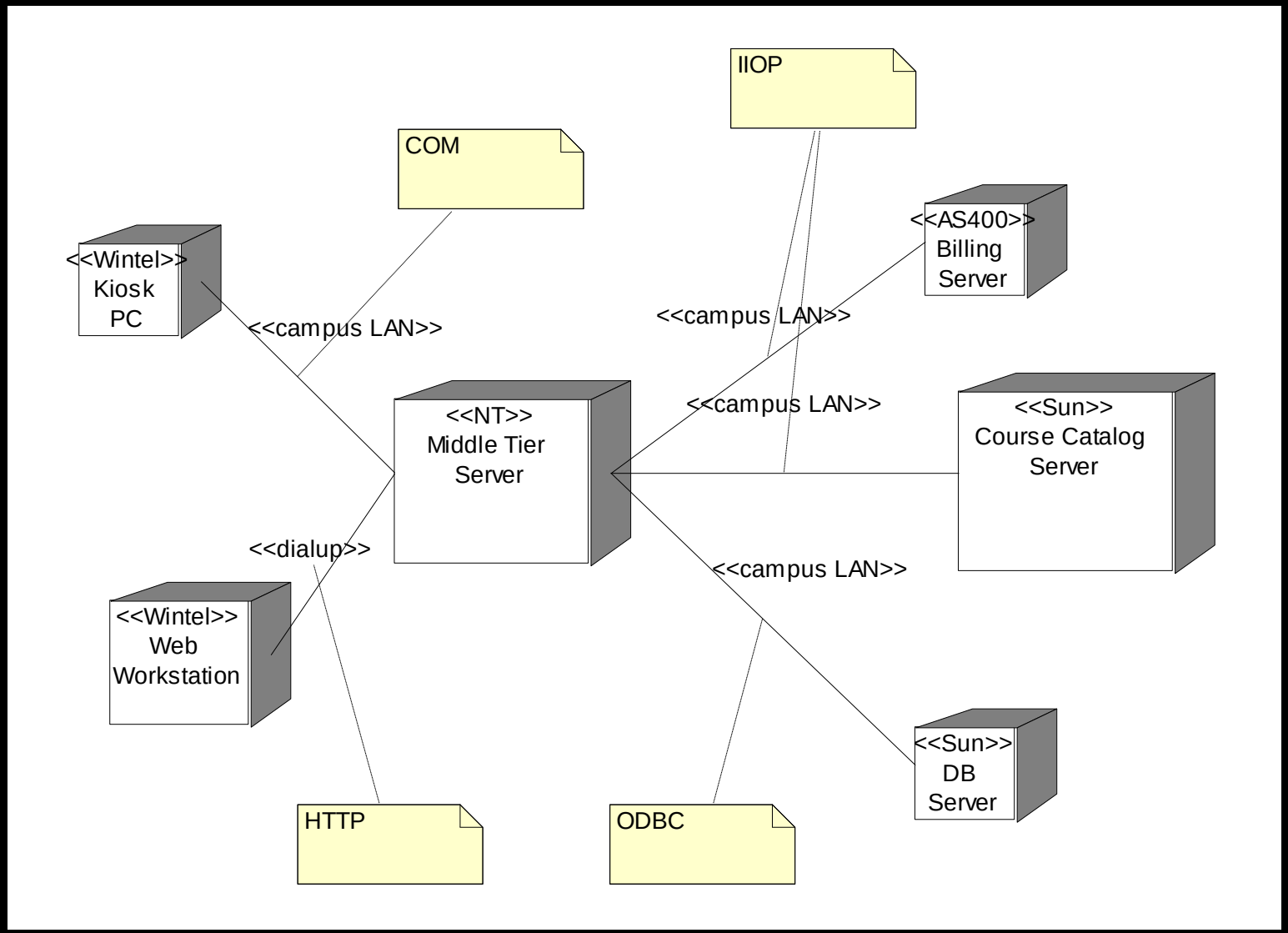
Example: Implementation View: Deployment Unit



Deployment View Representation

- Structure (network topology and element mapping)
  - Deployment diagram
    - Nodes
    - Elements mapped to nodes (from xxDe Mappings)
    - Connections
    - (At least) one deployment diagram per configuration

Example: Deployment View: Deployment Diagram



Example: Mapping

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File Edit View Insert Format Tools Table Window Help

3. Flow of events

3.1 Basic flow

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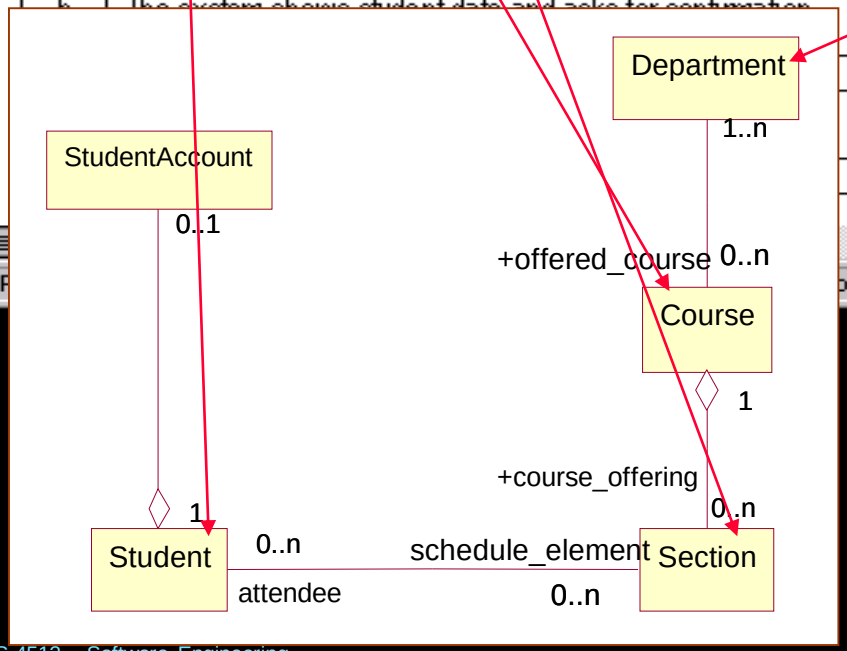
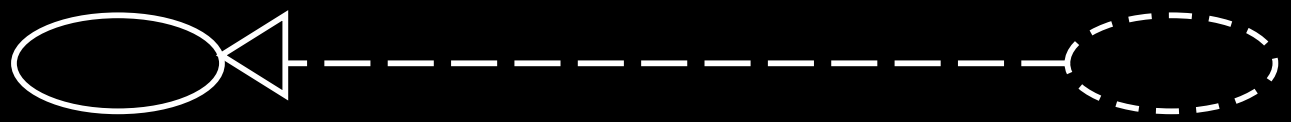


Use-Case Realizations

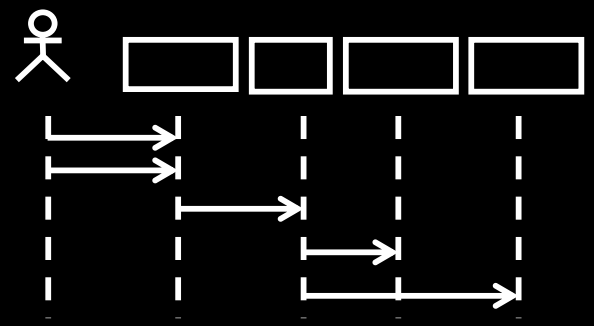
Use-Case View

Logical View (UCLo Mapping)

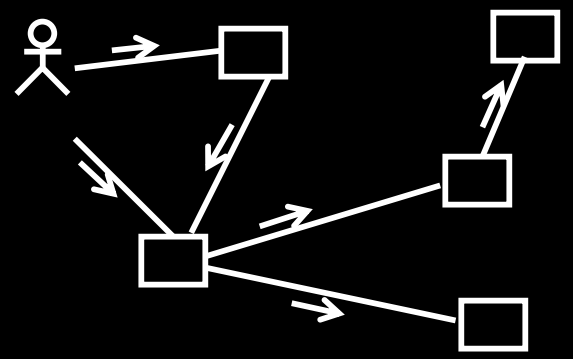


Use Case

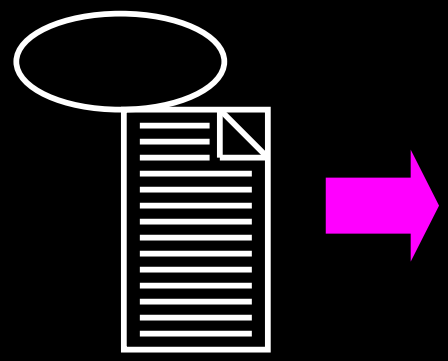
Use-Case Realization



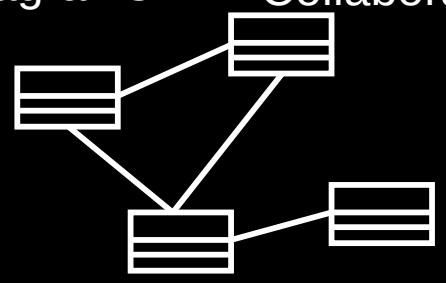
Sequence Diagrams



Collaboration Diagrams



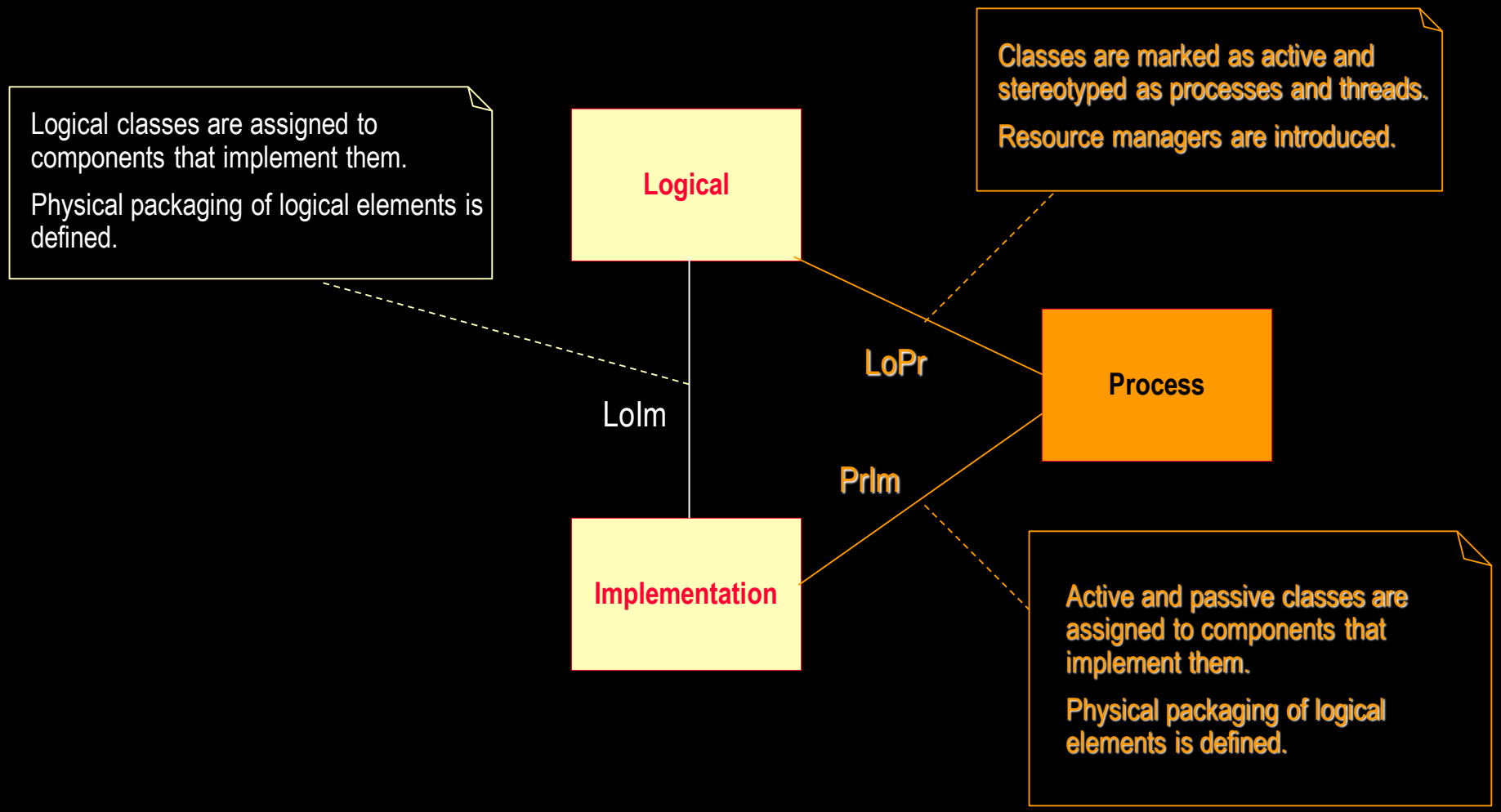
Use Case



Class Diagrams

Documented in the Logical View (separate diagrams)

Logical to Implementation View Mapping



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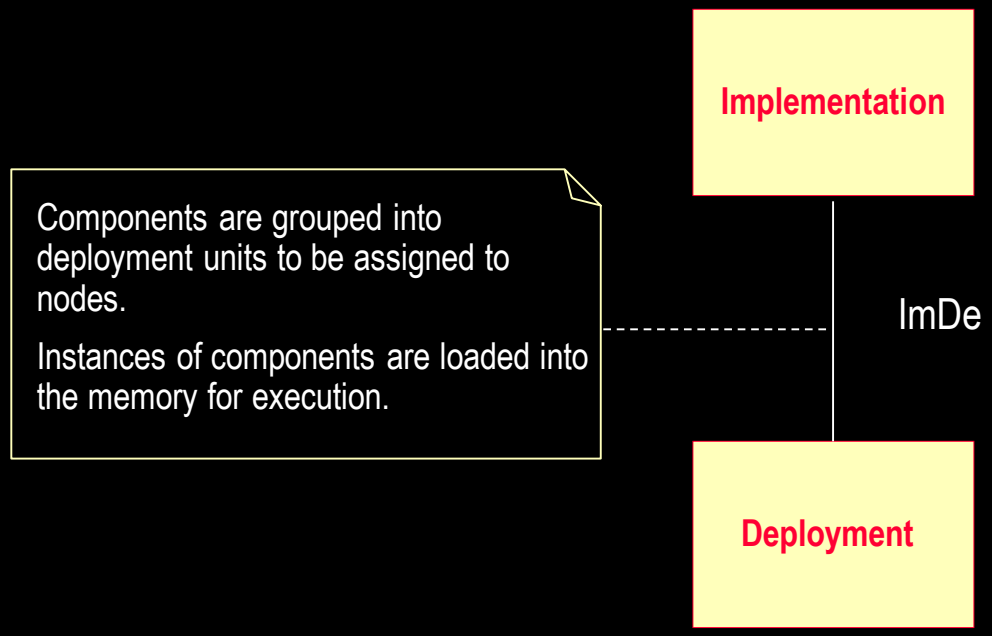
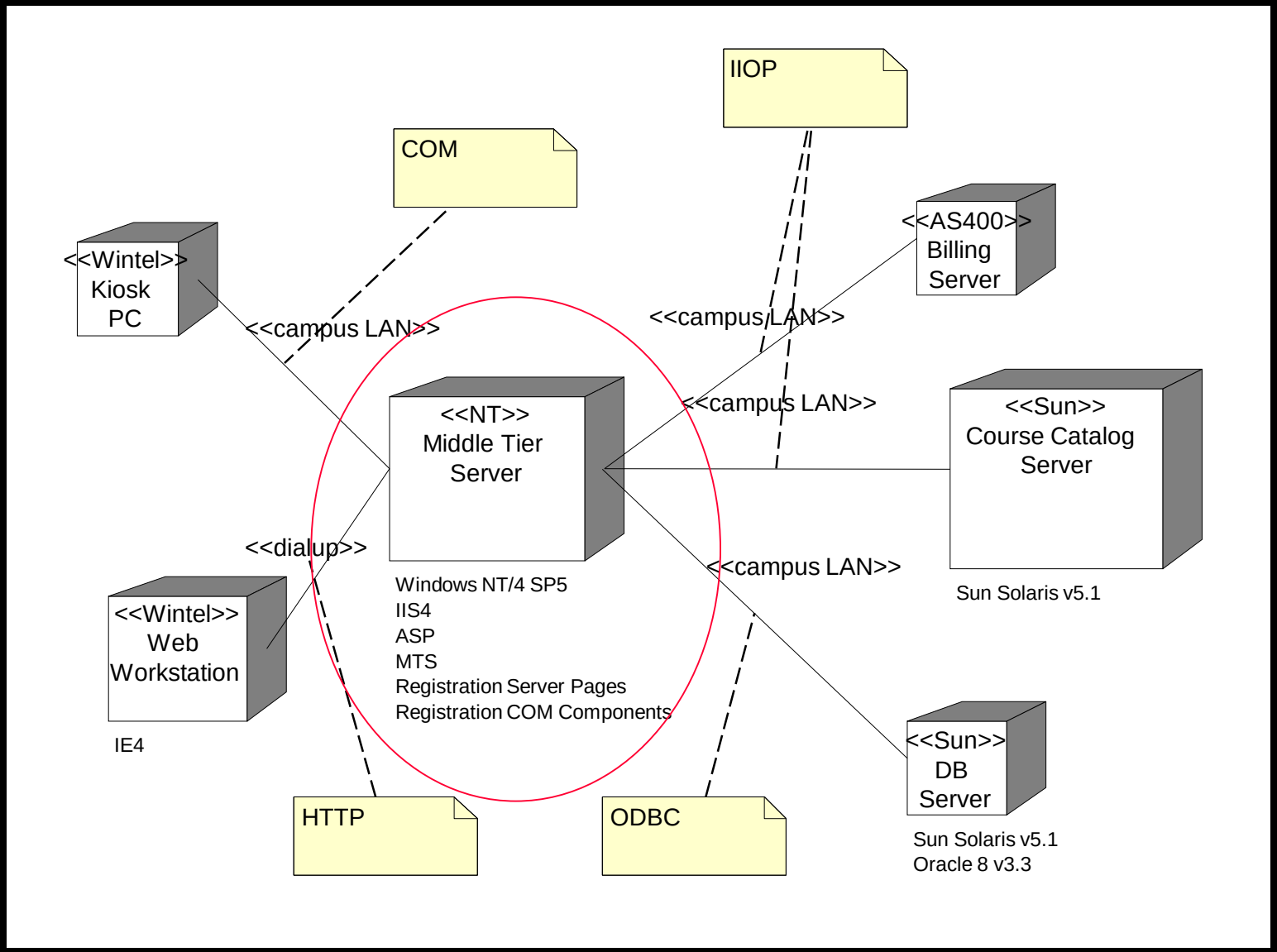
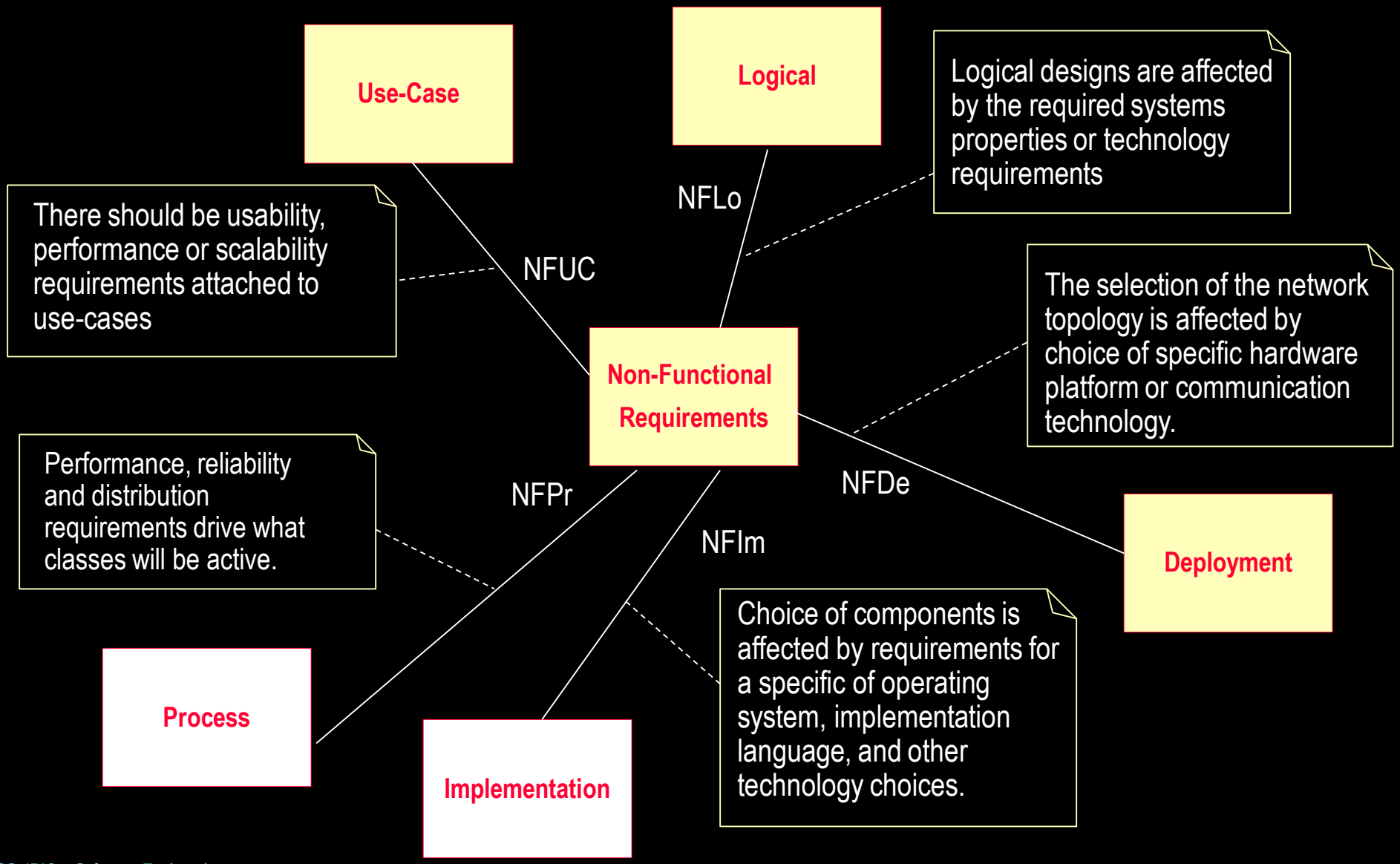
Implementation to Deployment View Mapping



Example: ImDe Mapping: Deployment Diagram



Non-Functional Requirements View to Other Views Mapping